IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Digital Object Identifier

Reliable Generalist Robot Policies for Manipulation: A Survey of Robustness Evaluation, Failure Detection, Recovery, and Continual Improvement

XIONGYULE ZHOU[1]
[1]Beijing Jiaotong University, Beijing 100044, China (e-mail: jensen.zhou21@gmail.com)
Corresponding author: Xiongyule Zhou (e-mail: jensen.zhou21@gmail.com).

ABSTRACT Generalist robot policies transfer visual and linguistic knowledge across manipulation tasks, but deployment reliability depends on how systems handle failures and learn from them. This structured critical survey reviews research published or posted through September 16, 2026, emphasizing work from 2022 onward. It develops a failure-conditioned evidence framework for evaluating reliability claims of generalist robot policies, linking failure occurrence, observable evidence, timely detection, recoverability, recovery outcome, and continual improvement. The framework connects robustness evaluation, uncertainty, replanning, human intervention, and recovery data by tracing interfaces among policies, monitors, recovery controllers, and learning processes. A reproducible corpus map distinguishes component, simulated, physical, and sustained-operation evidence. Five cross-paper reporting checks expose missing links between alarm timing and recovery outcomes while preserving event denominators and resource budgets. Three dependencies organize the synthesis: alarms are useful while recovery remains feasible; recovery outcomes depend on which failures are selected; and adaptation requires assessment of retained capabilities and monitoring validity. Usable-alarm recall is proposed as an operational metric for timely delivery on eligible recoverable events, alongside false-alarm burden. An illustrative reanalysis of published physical trials preserves the assumptions and uncertainty behind reported improvements. The proposed Minimum Reporting Framework for Recovery-Conditioned Reliability consolidates these requirements into an auditable report. Together, the framework and comparative synthesis support evaluation of operational completion and failure recurrence across the full reliability chain.

INDEX TERMS Continual learning, failure detection, generalist robot policies, manipulation, recovery, robustness evaluation, uncertainty, vision-language-action models.

I. INTRODUCTION

GENERALIST robot policies promise to replace repeated task-specific engineering with reusable visuomotor competence. Robotics Transformer 1 and 2 (RT-1 and RT-2), Octo, OpenVLA, and the π family demonstrate complementary routes toward this objective [1]–[7]. These models transfer information across tasks, scenes, or embodiments. Reliable execution additionally requires stable grasps, feasible contact, and verification that preceding actions succeeded.

A dropped object can leave the camera view, occupy an unseen pose, or obstruct the next action. The monitor must interpret this state, the recovery controller must restore continuation, and the learner must select useful experience. These dependencies organize the survey around failure and recovery. Throughout this article, reliability means sustained task completion within specified physical constraints, time and intervention budgets, and deployment conditions. Safety concerns unacceptable consequences and is related but not interchangeable: a robot can fail safely by stopping, and it can complete a task while violating a constraint.

Existing surveys already cover considerable ground. Foundation-model surveys discuss architectural transfer and embodied applications [8]–[10]; manipulation reviews connect planning with learning [11]. A particularly close survey by Kojima et al. [12] explicitly includes pre-deployment, pre-incident, and post-incident risk control, including recovery and improvement. The IEEE Access review by Kawaharazuka et al. [13] already addresses real-world deploy-

ment, platforms, datasets, and evaluation. Recent reviews additionally organize vision-language-action (VLA) architectures [14], adversarial safety [15], data engines [16], evaluation bottlenecks [17], long-horizon reinforcement learning [18], and continual robot learning [19]. A further lifecycle safety review explicitly includes both adversarial and non-adversarial risks [20]. The evaluation-centric review also supplies minimum reporting protocols and claim–metric alignment [17]; our framework specifically couples alarm delivery, recovery selection, and post-update recurrence. Table 1 identifies their overlap and the remaining interface questions.

Recent studies test shifted distributions and policy-generated failure states, while recovery demonstrations, language feedback, and experience-based updates are becoming explicit design choices [21]–[26]. Preliminary 2026 evidence is identified as such. Our three contributions follow the reliability chain in Fig. 1. First, a failure-conditioned evidence framework for reliability evaluation links failure occurrence, observable evidence, timely detection, recoverability, recovery outcome, and continual improvement. Second, monitor–recovery–learning interface analysis yields the proposed usable-alarm recall (UAR) metric and the *Proposed Minimum Reporting Framework for Recovery-Conditioned Reliability* in Table 8. These operational proposals preserve timing, selection rules, and event denominators. Third, the synthesis reconciles findings about scale, uncertainty, recovery data, and adaptation by identifying differences in experimental conditions.

The article answers five research questions (RQs). **RQ1:** Which shifts and execution mechanisms cause generalist manipulation policies to fail? **RQ2:** What do current benchmarks and physical evaluations actually establish about reliability? **RQ3:** When do uncertainty and failure monitors provide actionable evidence? **RQ4:** Which recovery mechanisms and human/data interfaces restore useful behavior? **RQ5:** Under what evaluation conditions does learning from failure constitute sustained improvement? These questions organize the methodology and comparative synthesis.

II. SURVEY METHODOLOGY

A. SCOPE, SOURCES, AND SEARCH

This is a structured scoping and critical survey, not a claim of exhaustive systematic retrieval or a statistical meta-analysis. Its search cutoff is September 16, 2026. The principal window is January 1, 2022 through that date; earlier work is included when it supplies a necessary basis for distribution shift, interactive imitation learning, safe recovery, calibration, or forgetting. The unit of inclusion is a research work, with a verified publication or explicitly identified preprint version. Closely related versions are consolidated; a later formal record does not create an additional independent experiment.

Discovery combined web-indexed primary-source searches with publisher, proceedings, arXiv, and Crossref retrieval. The September 16 extension attempted Scopus, Web of Science Core Collection, Google Scholar, IEEE Xplore, ACM Digital Library, PMLR/CoRL, RSS, and arXiv. Google Scholar returned ten parseable records on the retrieved first page. Scopus, Web of Science, native IEEE/ACM searches, and native arXiv search yielded no verifiable result set through the available access route; no database coverage or hit count is claimed for these attempts. PMLR and RSS were accessible through proceedings pages and site-restricted web discovery. The updated search log distinguishes exact queries, access probes, parsed result records, retained works, and unavailable counts. Merged web results are counted by retrieval batch, not attributed to individual queries or treated as native database exports.

The search families combined robot manipulation terms with (i) generalist policy or vision-language-action; (ii) robustness, distribution shift, generalization, benchmark, or real-world evaluation; (iii) failure, anomaly, uncertainty, confidence, or runtime monitoring; (iv) recovery, replanning, reset-free, correction, or human intervention; and (v) continual learning, adaptation, experience, or recovery demonstrations. Survey, review, and taxonomy variants were searched separately and refreshed for 2025–2026. Exact issued queries and source URLs are preserved in the search log. The protocol provides reusable Boolean strings; these are distinguished from the shorter queries actually accepted by the web search interface. Backward and forward title searches from close surveys and core studies provided targeted snowballing.

B. ELIGIBILITY AND SCREENING

An eligible core study must address manipulation or a directly transferable reliability mechanism and contain an inspectable technical account or primary abstract sufficient for the claim made here. Benchmarks qualify when their evaluation conditions are relevant even if they do not implement recovery. General policy and dataset papers qualify as context, not as direct evidence of failure detection. Classical driving or navigation studies are retained only as conceptual antecedents and are explicitly marked as such. Pure locomotion, text-only reasoning, unrelated anomaly detection, promotional summaries without research evidence, and records lacking reliable identity verification are excluded from technical conclusions.

Screening proceeded through title and abstract triage, identity/version verification, full-text or evidence-specific inspection, and structured extraction. Metadata disagreements were resolved in favor of the final publisher record for the cited publication, while an inspected author version was retained as a linked evidence source. For example, a conference meeting year and its proceedings publication year can differ; the bibliography uses the latter when that is the publisher's citation year. Unavailable full text limits claims to the primary abstract and excludes detailed numerical extraction. Inaccessible candidates remain visible in the exclusion log instead of receiving guessed metadata.

The resulting catalog contains 88 included research works: 56 verified formal publications and 32 cited preprints or publisher-labeled non-peer-reviewed manuscripts. Six works were added in the extension; formal FLARE and Diffusion

TABLE 1. Existing surveys versus the present survey: overlap and the remaining interface questions

Survey / cited version	Established coverage	Overlap with this article	Distinct emphasis here
Foundation-model reviews [8]–[10]	General-purpose robotics, architectures, learning and manipulation applications.	Transfer, embodied grounding, limitations.	Connect execution failures to timed responses and event-conditioned evidence.
Planning and learning [11]	Foundation-model manipulation through planning and learning perspectives.	Planning, policy learning, feedback.	Separate plan repair from physical state restoration and reset.
Physical risk control (2025) [12]	Pre-deployment, pre-incident and post-incident risk control, including recovery and improvement.	Direct lifecycle overlap; recovery is already covered.	Analyze monitor–recovery selection, deadlines and learning-data selection bias.
Real-world VLA review (2025) [13]	Full stack: architectures, hardware, data collection, augmentation and benchmarks.	Deployment constraints and evaluation.	Failure-event denominators and sustained autonomous operation.
VLA architecture survey (2026) [14]	VLA foundations, methods, embodied applications and challenges.	Generalist policy mechanisms.	Compare architectures by observable signals and controllable recovery interfaces.
VLA safety (2026 preprint) [15]	Attacks, defenses, evaluation and deployment; training/inference timing.	Runtime safety and distribution changes.	Unintentional execution failures and task restoration under budgets.
Data engines (2026 preprint) [16]	Datasets, benchmarks and data engines; fidelity and cost.	Coverage and evaluation infrastructure.	Provenance of naturally induced, injected and reconstructed failure states.
Evaluation-centric review (2026 preprint) [17]	Benchmark validity, reporting protocols and claim–metric alignment.	Evaluation and minimum-reporting overlap.	Couple operational metrics to available repair and post-update retention.
Long-horizon RL (2026) [18]	Scoping review of reinforcement learning for extended manipulation.	Long horizons, learning and evaluation gaps.	Include non-RL VLA monitors, intervention channels and failure-conditioned comparison.
Lifecycle VLA safety (2026 preprint) [20]	Data, training and deployment; adversarial and non-adversarial risks.	Substantial lifecycle and reliability overlap.	Event-conditioned task restoration, timing and operational denominators.
Continual robot learning (2026 preprint) [19]	Task formulations, methods, benchmarks and retention in foundation-model robotics.	Continual adaptation and forgetting.	Effects of intervention selection and monitor drift on improvement claims.
This survey	Robustness, detection, recovery, human/data interfaces and continual improvement.	Builds on, rather than claims priority over, lifecycle reviews.	UAR; recovery-conditioned denominators; proposed Table 8; monitor validity after updates.

Coverage statements identify organizing emphasis, not an assertion that an earlier survey never mentions another topic. Publication status refers to the specific version in the bibliography.

Policy versions replaced earlier citations without adding independent works. These counts describe the included corpus. The screening ledger retains candidate and version decisions, including potentially relevant records not selected for this bounded critical synthesis. It is not an exhaustive eligibility census. The author retains responsibility for final study selection, verification, coding, and interpretation. Independent dual screening and inter-rater agreement analysis were not performed. The bibliographic refresh checked all records against primary metadata where accessible; restricted refreshes retain their previously verified sources and explicit limitations.

C. EXTRACTION AND SYNTHESIS

The database stores title, ordered authors, year, venue, DOI or preprint identifier, publication status, verification URLs, access date, and evidence availability. Technical extraction records the policy or component studied, disturbance type, signal and supervision, recovery target, evaluation substrate, and principal inferential limit. Quantities are included only when a source table and denominator can be located. Missing information is marked as not established by the evidence extracted, rather than encoded as a negative result.

Synthesis follows RQ1–RQ5 across eight main-text tables. The reproducible evidence map in Fig. 6 uses explicit domain–substrate pairs from 51 full-text-coded studies; 34 contextual works and three records with restricted full-text extraction are kept outside that map. Multiple codes are permitted, so cell counts are not additive. Physical evidence can include retrospective monitoring of physical rollouts; it does not imply an online repair trial. Component evidence covers isolated reasoning or recorded-outcome tests. Sustained operation requires repeated physical execution with duration and quantified reset or assistance frequency/cost; a long task alone does not qualify. Continual/adaptive improvement includes evaluated updates, with retention distinguished in the extraction record. Unassigned cells mean no positive code in this extraction, not absence of capability. Five purposively selected studies receive a separate field-level Table 8 comparison (Supplementary Table S1). No cross-task effectiveness estimate is pooled.

III. GENERALIST ROBOT POLICIES AND THEIR RELIABILITY INTERFACES

A. GENERALIST IS A TRANSFER CLAIM

A generalist policy is intended to reuse a learned controller or representation across multiple tasks, objects, environments, or embodiments. Its scope is the declared transfer distribution. VLA models are a major subset, coupling visual and language-conditioned representations to robot actions; they do not exhaust generalist policy design. Diffusion Policy and action-chunking approaches illustrate useful controller mechanisms even when a particular experiment is task-specific [27], [28]. Table 2 compares architectures by the interfaces they expose to monitoring and recovery rather than parameter count.

Large and diverse training corpora expand nominal support. Open X-Embodiment, BridgeData V2, and DROID contribute different combinations of embodiment, task, and scene diversity [29]–[31]. Successful demonstrations may omit overturned objects, near-collisions, stalled contact, and other policy-induced states. Recovery coverage depends on sampling these consequences as well as nominal task diversity.

B. REPRESENTATIONS CHANGE FAILURE OBSERVABILITY

Discrete action prediction provides convenient token-level quantities but a token probability is conditional on the model's training objective and representation. Interpreting it as task-success probability requires outcome calibration. Continuous diffusion or flow policies model multimodal actions, so disagreement between valid samples can reflect several acceptable strategies rather than epistemic uncertainty. Conversely, a single coherent but mistaken mode can produce confident failure. These issues motivate monitors that observe temporal consistency and task progress in addition to internal policy scores [32].

Action chunks reduce the frequency of expensive model inference and may support smooth coordinated motion, but their execution horizon also determines how quickly new visual evidence changes control. The relevant latency includes the portion of an already issued chunk that cannot be canceled. Elastic Queries Reinforcement Learning makes inference budget and action-chunk length state-dependent, illustrating a concrete execution-scheduling response to this issue [33]. Scheduling efficiency and deadline satisfaction nevertheless remain different evaluation targets. Architectures that support language, subgoal images, or other conditioning can provide recovery interfaces, as illustrated by the steerability studied in $\pi_{0.7}$ [7]. Such interfaces still require a supervisor to identify a physically appropriate instruction and a controller capable of executing it.

High-level language planning and low-level policy execution also have different semantics. SayCan grounds candidate skills in affordance estimates [34]; an available skill is not a guarantee that its preconditions remain true after a perturbation. Recovery must therefore account for the boundary between task-level symbolic commitments and continuous physical state. A generalist architecture is most useful to this survey when it exposes an observable state, a controllable intervention, or a learning update that can be evaluated independently.

IV. RELIABILITY AND DISTRIBUTION SHIFT

A. EXTERNAL AND POLICY-INDUCED SHIFTS

External shifts change what the system encounters: lighting, clutter, object geometry, language, calibration, embodiment, or dynamics. Policy-induced shifts arise when an imperfect action changes the subsequent state distribution. Policy-induced changes connect manipulation to the covariate shift underlying dataset aggregation [35]: failed insertion or displaced objects alter subsequent control requirements even in visually familiar scenes.

The STAR-Gen taxonomy distinguishes visual, semantic, and behavioral generalization and motivates factorized evaluation [36]. These axes are useful but interact during failure. Moving an object can change both its visual appearance and the action required; a new embodiment changes sensor geometry as well as control affordances. Consequently, controlled single-factor tests answer a different question from combined-shift deployment. Together they characterize isolated sensitivities and interacting shifts. Geometric Red-Teaming searches plausible object deformations and tests targeted policy repair, linking failure discovery to an explicit repair distribution [37].

B. RELIABILITY IS CONDITIONAL ON AN OPERATING ENVELOPE

Let g denote a goal and let $\mathcal{C}$ contain declared physical and operational constraints. Reliability claims must specify an operating envelope $\mathcal{E}$, a wall-clock budget B, and allowed assistance H. A useful target is

$$P(S_{\text{auto}} \mid \mathcal{E}, B, H = 0), \tag{1}$$

where S_{auto} means that the original goal is reached within the budget while satisfying $\mathcal{C}$ and without human takeover. This reporting definition fixes the operating conditions and assistance allowance when comparing success rates.

Robustness concerns behavior under specified disturbances; uncertainty concerns what a model or estimator expresses about its predictions; calibration concerns agreement between a predicted quantity and its empirical frequency. None is a synonym for reliability. Calibration on nominal data can coexist with poor shifted performance [38], [39]. A safety filter can satisfy a constraint by refusing action while making no progress toward the task. Reliability evaluation must keep these outcomes visible rather than compress them into one undocumented score.

C. THE RECOVERY DISTRIBUTION IS ENDOGENOUS

Failure states depend on the policy, alarm threshold, human behavior, and prior recoveries. Generic perturbations can improve coverage while missing naturally occurring errors. Policy-generated recovery benchmarks test a specified failure distribution [23]; transfer to other policies or deployment

TABLE 2. Policy families compared by reliability-relevant interfaces

Family / examples	Action or conditioning interface	Useful monitoring handle	Reliability limitation
Tokenized control [1], [2], [4]	Discrete robot-action prediction with visual/language context.	Token distributions and observed task progress.	Token likelihood is not calibrated physical-success probability.
Generalist diffusion [3]	Denoising-based action generation; task and embodiment conditioning.	Samples, observations and action consistency.	Multimodal valid actions can create disagreement without failure.
Diffusion / action chunks [27], [28]	Predicted action sequences executed through a control loop.	Temporal consistency; chunk boundaries; feedback timestamps.	Executed horizon and cancellation delay limit response timing.
Flow-based VLA [5], [6]	Continuous action generation coupled to broad visual/language representations.	External monitor or learned input/output score.	Broad nominal transfer does not certify post-failure support.
Steerable generalist (preprint) [7]	Language plus contextual information, including strategy and subgoal cues.	Conditioning can express a desired corrective behavior.	A steerable interface still needs grounded diagnosis and feasible control.
Language skill planning [34]	Language candidates combined with skill affordance estimates.	Preconditions, skill outcomes and plan progression.	A valid plan cannot repair an unavailable low-level skill.

Architectures are not ranked. Monitoring handles are this survey's analytical interpretation of interfaces, not claims that each policy ships with a validated failure detector.

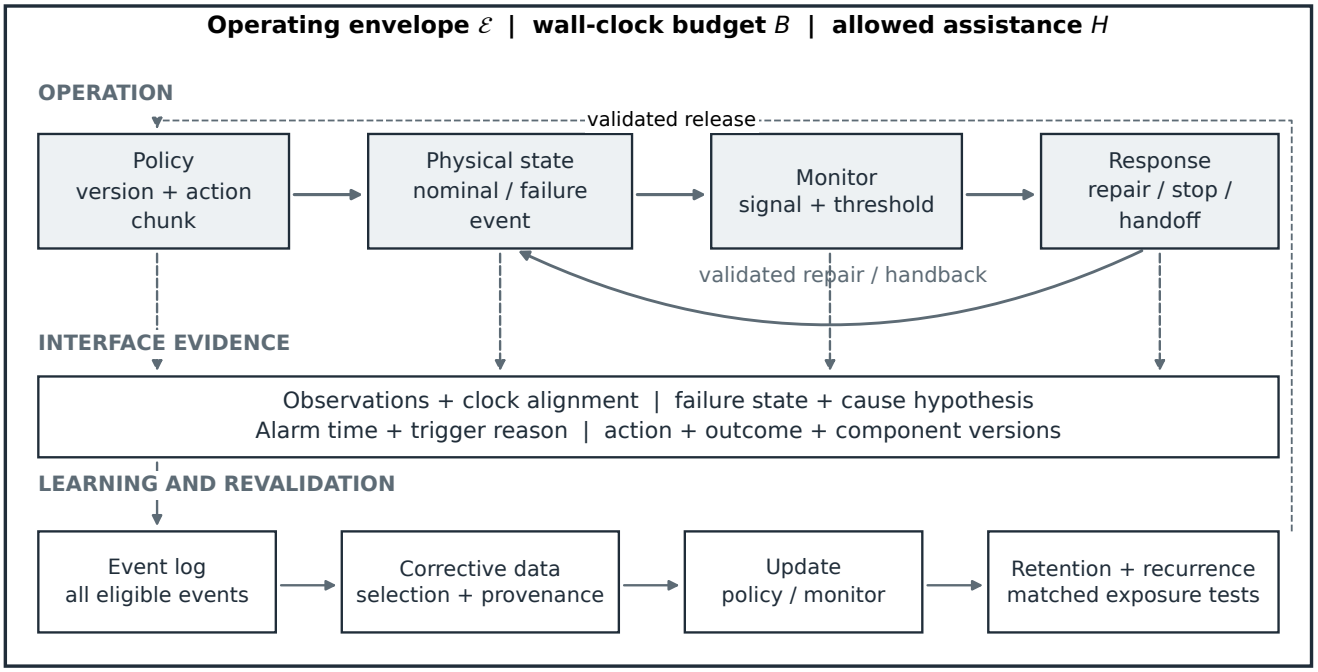


FIGURE 1. Reliability system architecture under envelope $\mathcal{E}$, budget B, and assistance allowance H. Interface evidence connects execution, response, and revalidation; Table 8 specifies the proposed minimum reporting fields.

regimes requires separate evaluation. We distinguish nominal states, disturbed initial states, and post-failure states throughout the comparison.

V. ROBUSTNESS EVALUATION AND REAL-WORLD EVIDENCE

A. WHAT BENCHMARK FAMILIES MEASURE

Table 3 separates benchmark substrates and reliability questions. CALVIN studies language-conditioned long-horizon manipulation, whereas LIBERO was designed to examine knowledge transfer in lifelong robot learning [40], [41]. Reporting a single multitask LIBERO score does not, by itself, execute its lifelong-learning evaluation. RoboCasa and RoboCasa365 broaden simulated household settings and tasks [42], [43]. Their breadth supports controlled testing; contact fidelity and maintenance require hardware evidence.

THE COLOSSEUM, LIBERO-PRO, and LIBERO-X foreground robustness under controlled changes [21], [22], [44]. Their importance is methodological: repeated nominal task success can mask dependence on initial-state, visual, or instruction regularities. However, shifted initialization and failure recovery remain different tests. LIBERO-RECOVER explicitly starts from policy-generated failure states and evaluates recovery execution and recovery training in simulation [23]. Its four levels distinguish retry, action adaptation, object-state recovery, and environmental recovery. These failures are generated by the evaluated policies in simulation.

Other benchmarks test components. ManipBench probes low-level manipulation understanding and separately tests transfer through hardware keypoint control [45]. These component tests reveal reasoning deficiencies; closed-loop tests must additionally address sensor latency and contact uncertainty. FailBench extends this distinction to outcome judgment across recorded manipulation attempts from physical and simulated sources [46]. Its cross-domain findings caution that failure-specific fine-tuning need not improve a visual-language evaluator on a new source. Recorded physical videos constitute an evaluator test, not an online physical recovery experiment.

B. SIMULATION AND PHYSICAL EVALUATION ANSWER COMPLEMENTARY QUESTIONS

The Simpler evaluation work compares manipulation policies in simulation with real-world behavior and studies how simulation can support policy assessment [47]. Transfer depends on represented geometry, appearance, embodiment, and dynamics. Simulation supports reproducible interventions and counterfactual starts; hardware tests address synchronization, friction, compliance, drift, and maintenance.

AutoEval addresses the infrastructure cost of autonomous physical evaluation [48]. Its reset and success-assessment components are themselves parts of the measurement process. If the evaluator misclassifies success or restores states unevenly, the resulting policy score inherits that error. Active factor-based physical evaluation uses an adaptive allocation strategy to select informative configurations [49]. Adaptive allocation targets informative configurations; estimating performance under a fixed deployment distribution additionally requires accounting for the sampling rule.

C. DENOMINATORS, BUDGETS, AND LEAKAGE

Distinguish success per attempt, per episode with declared retries, and per operating period. Retries can increase completion while reducing throughput; resets add assistance. Figure 2 separates these measurement levels. An evaluation should report the number of initiated episodes, encountered failure events, alarms, recovery attempts, human takeovers, and completed goals, together with elapsed time and reset cost.

Train–test separation must also apply to failure states, not only task names. Reusing nearly identical disturbances or recovery templates across training and evaluation can measure memorized repair rather than robustness. The recent benchmark audit [50] raises broader questions about shortcuts, statistical significance, and benchmark reuse. Here the practical implication is to retain independent seeds, object instances, initial and failure states, and preferably policy families when constructing evaluation splits. Tests should identify which of these axes were actually held out.

Evaluation quality is not captured by sample size alone. Repeated trials from one arrangement may be strongly correlated; wide task coverage with one trial per task may poorly estimate each task's failure probability. Both the experimental unit and allocation need disclosure. Confidence intervals should match that unit, with hierarchical or cluster-aware analysis when trials share scenes, sessions, or operators. A binomial interval is defensible only under a corresponding trial model, not as a universal remedy for heterogeneous robot experiments.

VI. FAILURE TAXONOMY

A. SEPARATE CAUSE, SYMPTOM, CONSEQUENCE, AND RESPONSE

Record the cause of a deviation, its observable symptom, the resulting task or constraint violation, and the response that may restore feasibility. ''Low confidence'' is a signal, not a physical failure cause. ''Dropped object'' is an event and state change, but it may follow poor geometry estimation, insufficient grip, unexpected mass, or coordination error. Treating these labels as interchangeable makes detector and recovery comparisons ambiguous.

Table 4 and Fig. 3 organize failures across perceptual grounding, instruction or planning mismatch, geometric execution, contact and dynamics, temporal coordination, progress and termination, and infrastructure or supervision. These overlapping coding categories can coexist within an episode; their prevalence is not estimated here. A causal label requires more evidence than a visible symptom, particularly for vision-only systems.

B. RECOVERABILITY IS RELATIONAL

Whether a state is recoverable depends on available actions, resources, constraints, and time. An object outside a camera view may be recoverable through active perception, unrecoverable with a fixed view, or recoverable only through a human. A spilled hazardous substance may make immediate stopping the correct response even if a nominal goal remains physically attainable. Therefore, a taxonomy should attach a target and an allowed action set to the word recovery.

We distinguish *retry*, which repeats or locally modifies an action without restoring a broader prerequisite; *state restoration*, which reconstructs useful object or robot conditions; *replanning*, which changes the remaining route to the goal; and *safe termination or handoff*, which contains consequences without claiming autonomous completion. An *environment reset* prepares a new trial or training episode. It may resemble restoration physically, but it does not necessarily continue the original task. This separation is especially important when

TABLE 3. Benchmark and evaluation comparison: what is tested and what remains outside the claim

Benchmark / study	Substrate and unit	Reliability question	Boundary for recovery claims
CALVIN [40]	Simulated language-conditioned task sequences.	Long-horizon composition and transfer.	Sequence completion alone does not isolate failure recovery.
LIBERO [41]	Simulated task suites and learning schedules.	Knowledge transfer for lifelong robot learning.	A static multitask score is not a sequential-retention study.
RoboCasa / RoboCasa365 [42], [43]	Simulated household tasks and environments.	Breadth of manipulation training and evaluation.	Scene diversity does not establish physical contact fidelity.
THE COLOSSEUM [44]	Simulated controlled perturbations.	Generalization under varied task conditions.	Disturbed conditions differ from naturally induced failure states.
Simpler evaluation [47]	Simulation with physical-policy comparisons.	When simulated evaluation reflects real policy behavior.	Transfer is conditional on represented task and embodiment factors.
STAR-Gen [36]	Taxonomy with physical manipulation studies.	Visual / semantic / behavioral generalization.	Factor-based tests do not automatically measure recovery cost.
AutoEval [48]	Physical autonomous evaluation pipeline.	Scalable trials with reset and outcome assessment.	Evaluator errors and maintenance affect the measured policy score.
LIBERO-PRO / LIBERO-X [21], [22]	Simulated changed evaluation conditions.	Robustness beyond nominal benchmark regularities.	Recovery must be evaluated explicitly rather than inferred from robustness.
LIBERO-RECOVER (preprint) [23]	Simulated execution from policy-generated failures; recovery training.	Retry, adaptation, object-state and environment recovery.	Failure-generating policies define the test distribution; not hardware trials.
ManipBench / FailBench [45], [46]	Component tests; ManipBench also validates hardware control.	Low-level understanding or cross-domain success judgment.	Recorded physical evidence is not an online recovery experiment.
Active physical evaluation (preprint) [49]	Adaptive selection of real-world factor configurations.	Efficient discovery/modeling of policy performance variation.	Adaptive sample averages need not estimate a fixed deployment average.
Benchmark audit (preprint) [50]	Analysis of manipulation benchmarking practices.	Shortcuts, statistical evidence and reuse effects.	An audit is not a new universal recovery leaderboard.

Rows may combine related benchmarks to keep their evaluation role clear. Simulated, physical and component-level evidence are not pooled. The 2026 preprints are preliminary.

comparing autonomous learning infrastructure with recovery during deployment.

C. LABELS MUST MATCH THE DECISION TIMESCALE

Episode-level labels identify eventual failure; actionable warnings and intervention deadlines require temporal annotation. Timestamp-level labels and weak supervision address a different problem, as illustrated by FailureSpot [51]. Propagating a terminal failure label to every earlier frame would mark initially correct behavior as failure. For monitor evaluation, annotation should distinguish onset, observable evidence, an intervention deadline when definable, and terminal outcome. Uncertain or interval-valued onset labels are preferable to artificial precision.

VII. FAILURE DETECTION AND UNCERTAINTY

A. SIGNALS HAVE DIFFERENT MEANINGS

Table 5 compares monitors by input evidence, supervision, output, and inferential limit. Deep ensembles estimate variation across learned predictors [52]; calibration methods map scores to empirical correctness under a specified distribution [38]. Using either for generative control requires calibrating sampled-action dispersion against physical failure events. Task ambiguity can yield several correct actions, while a biased model can be consistently wrong. A detector should therefore be validated against the event it is intended to trigger, rather than against a convenient surrogate alone.

Sentinel combines temporal action consistency with semantic progress monitoring [32]. Its distinction between erratic and confidently wrong behavior shows why no single uncertainty signal covers all failure mechanisms. FAIL-Detect instead learns or extracts scalar signals from successful policy data and uses a conformal framework for sequential out-of-distribution detection [53]. Success-only training reduces dependence on enumerating known failure examples, but a novel state and an unsuccessful state are not identical concepts. Score utility depends on alignment with intervention-worthy deviations. SAFE evaluates VLA-feature-based detection on unseen tasks and reports false-positive rates and normalized detection time [54]; these quantities still require pairing with a feasible response to measure UAR.

Vision-language models provide semantic judgments and explanations. Success detectors address goal satisfaction [55]; AHA studies failure detection and reasoning with generated failure examples [56]. These outputs can guide response selection; causal diagnosis additionally requires evidence that distinguishes competing explanations. An absent object can reflect a drop, occlusion, or intentional movement; physical and temporal evidence must distinguish these alternatives.

B. PHYSICAL AND MULTIMODAL FAILURE SIGNALS

Vision and VLM judgments describe object relations and task progress, while physical sensing exposes interaction at the robot–object interface. Yang et al. [57] compare anomaly detectors using force, torque, velocity, tactile, and kinematic

	Component	Simulation	Physical episode	Sustained operation
Policy evaluation	Representation / action checks on fixed inputs	Completion under controlled disturbances	Hardware completion in the tested envelope	Throughput and failure rate over declared time
Failure detection	Label / score agreement on supplied observations	Alarm timing against simulated event labels	Delivered alarms and contact observability	Missed / false alarms per operating exposure
Recovery	Repair proposal or precondition consistency	Closed-loop repair from specified states	Restoration and original-goal continuation	Recurrence, downtime and repair-induced faults
Human assistance	Quality of proposed queries or corrections	Effect of a specified assistance policy	Measured operator effort and response delay	Staffing, maintenance and accumulated burden
Long-term reliability	Version / calibration consistency checks	Retention across a simulated update stream	Retention on repeated hardware task tests	Reliability across updates, wear and maintenance

Claims require direct evidence on the named substrate; columns are not a ranking.

FIGURE 2. Evidence–claim matrix. Each cell identifies a claim requiring direct measurement on the stated substrate; no ordering of scientific quality or universal transfer between columns is assumed.

observations in a physical human–robot kitting task. This evidence supports treating sensor combinations and anomaly timing as evaluation variables. Wrist force/torque indicates unexpected loading; tactile observations constrain local contact state; and proprioceptive position or velocity supplies execution context. Their meaning depends on the task phase: expected insertion force and unintended obstruction can produce similar instantaneous loads. Combining temporal and multimodal evidence therefore complements semantic monitoring rather than replacing it.

Motor current can provide another contact-sensitive channel when dedicated external sensors are unavailable. Park et al. [58] evaluate one-class support-vector-machine and autoencoder detectors using current measurements with a robot dynamics model, trained on collision-free motion. The demonstrated target is collision detection, with model and embodiment dependence; extending it to VLA task-failure diagnosis requires paired event annotations. For early warning, the useful question is whether a modality advances the first informative observation enough to permit a feasible response. Synchronization, filtering, inference, and actuator cancellation must all be included in the timing account. We propose matched modality-ablation tests reporting usable-alarm recall, false-alarm burden, and original-goal continuation under the same recovery controller, including sensor dropout and expected contact as separate conditions.

C. STATISTICAL GUARANTEES HAVE A SCOPE

Conformal prediction can construct calibrated sets or alarms under stated sampling and score assumptions [39]. A sequential deployment also raises a multiple-testing issue: trajectory-level false-alarm control requires accounting for repeated decisions in addition to marginal error at one time. A paper's actual sequential construction must therefore be distinguished from applying a static threshold repeatedly. Moreover, calibration assumptions may cease to hold after a camera change, policy update, or new recovery controller changes the observation distribution.

KnowNo applies uncertainty alignment to language-based planning and requests help when appropriate [59]. Its decision target differs from motor-level failure detection: ambiguity among plausible plans is not the same event as impending contact violation. KnowLoop examines uncertainty-based monitoring in closed-loop language planning [60], and latent safety filters consider avoiding out-of-distribution failures through action constraints [61]. Such complementary approaches belong at different interfaces. A guarantee about plan selection or a learned latent constraint should not be presented as a theorem of physical manipulation safety.

D. ACTIONABLE ALARMS REQUIRE TIME AND A VIABLE RESPONSE

Let t_{onset} denote failure onset, t_{obs} the first informative observation available to the monitor, and t_a the emitted-alarm timestamp. Let L include all remaining delay after alarm

TABLE 4. Failure taxonomy: causal hypotheses, observations and response targets

Mechanism / example	Observable symptom	Candidate response target	Learning/evaluation implication
Perception / grounding	Wrong object, lost object, occlusion or inconsistent pose.	Reobserve, change viewpoint or re-ground the target.	Distinguish sensor evidence from a guessed physical cause.
Instruction / plan mismatch	Wrong subgoal, unmet precondition or impossible skill order.	Clarify the goal or revise remaining skills.	Score plan validity separately from motor execution.
Geometric execution	Missed grasp, insertion misalignment or reachability failure.	Retry with changed geometry or restore a useful pose.	Hold out failure poses as well as nominal initial poses.
Contact / dynamics	Slip, unstable support, jamming or unintended displacement.	Release safely, regrasp or restore object/support state.	Visual correctness may not reveal stable contact.
Timing / coordination	Stale observation, uncanceled chunk or bimanual mismatch.	Interrupt and resynchronize within a feasible deadline.	Include communication and actuation in latency.
Progress / termination	Stall, loop, premature success or endless retry.	Change strategy, verify completion or terminate.	Use time budgets and explicit stop/retry rules.
Infrastructure / supervision	Calibration drift, sensor dropout or erroneous evaluator label.	Restore sensing/calibration or obtain assistance.	Version evaluator and hardware conditions with the policy.
Repair-induced deviation	New obstruction, unsafe return or recurrence after handback.	Monitor the repair and verify a resumable state.	Count recovery failures and subsequent recurrence.

Original synthesis informed by the mechanism and evaluation distinctions in the cited detection, recovery and benchmark literature. Categories overlap; no prevalence counts or exclusive causal assignments are claimed.

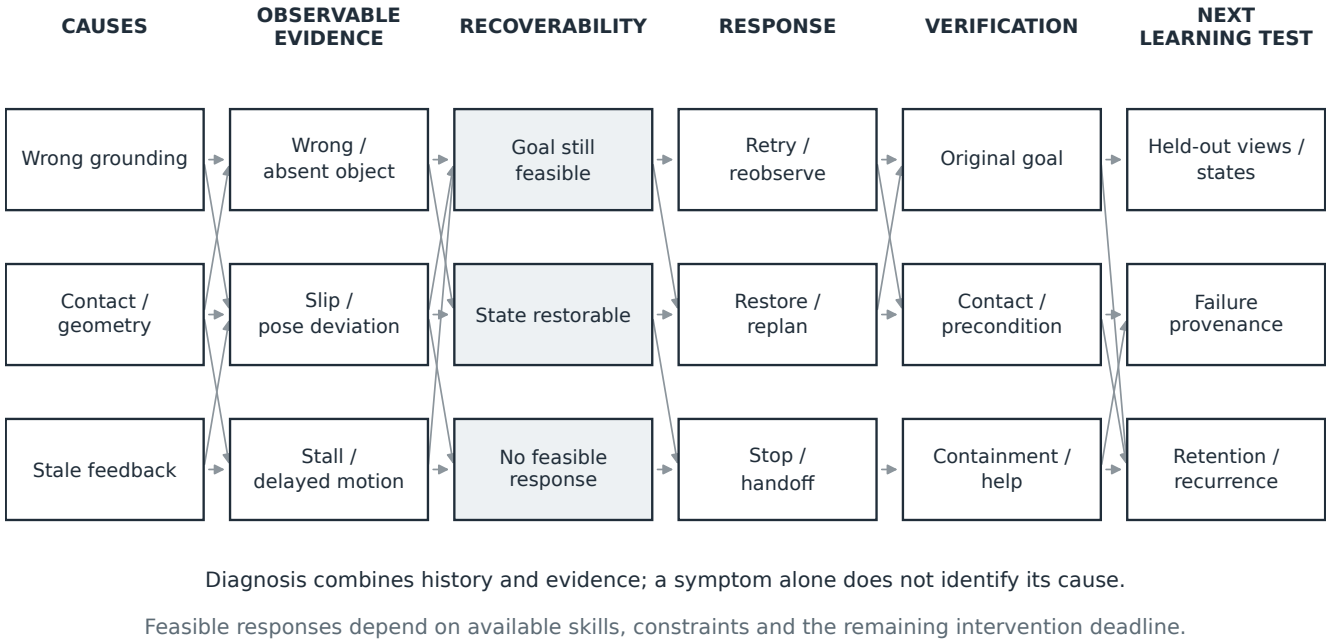


FIGURE 3. Many-to-many failure analysis. Illustrative links separate causes, observations, recoverability, responses, verification targets, and subsequent learning tests. Diagnosis combines evidence with execution history.

emission: any additional sensing or planning, communication, action cancellation, and actuation needed to deliver the response. The interval $t_a - t_{onset}$ accounts for observation and detection latency before the alarm and is not counted again in L. Let t_d be the last time an allowed response can preserve the specified recovery target. An alarm is operationally timely only if

$$t_a + L < t_d. \tag{2}$$

The strict inequality is a conservative convention; a discrete controller may use an explicitly defined inclusive deadline. Figure 4 illustrates why earlier recognition is useful only when the response can be delivered in time. The deadline is task- and controller-dependent and may be uncertain. Its existence cannot simply be inferred from the time of terminal failure.

We propose the operational metric *usable-alarm recall* (UAR)

$$UAR = \frac{\#\{\text{recoverable events with timely delivery}\}}{\#\{\text{all eligible recoverable failure events}\}}, \tag{3}$$

alongside the protocol defining recoverability and deadlines.

If no eligible recoverable events occur, UAR is undefined and should be reported as not applicable with a zero event count. The denominator must include missed alarms, and evaluation should also report false alarms per operating hour and their interruption cost. UAR is a proposal of this survey, not an established benchmark metric. It complements receiver-operating-characteristic analysis by measuring whether delivered alarms preserve the declared recovery option. Where deadlines cannot be annotated reliably, outcome-versus-delay interventions can estimate an empirical response curve, with appropriate limits on physical testing.

VIII. RECOVERY AND REPLANNING

A. RECOVERY TARGETS DETERMINE CONTROLLER REQUIREMENTS

Table 6 distinguishes returning to a safe set, reconstructing a task prerequisite, correcting local execution, and changing a task plan. Recovery RL separates a task policy from a controller that returns behavior toward learned safe regions [62]. Evidence includes simulated contact-rich tasks and physical obstacle avoidance. Transfer to contact-rich hardware requires further testing; the separation of task reward from constraint avoidance remains relevant.

Plan repair uses semantic state and skill abstractions. REFLECT summarizes execution evidence for failure explanation and correction [63], while RePLan and DoReMi investigate replanning or detecting plan–execution mismatch [64], [65]. These methods are most naturally suited to failures expressible through task relations, violated preconditions, or available skills. If the required corrective motion lies outside the skill library, a correct diagnosis cannot create the missing controller. Conversely, a local visuomotor repair may succeed without revising the high-level plan. VLCP studies another repair level: repeatedly rewriting a control function after new observations [66]. Its simulated evaluation supports revisable control interfaces; physical actuation and safe code execution require separate validation.

RACER connects richer language feedback to a conditioned recovery policy [24]. Its interpretation–action interface depends on distinctions represented in training data and grounded in the current scene. The interface must specify the target state, permitted action, and return condition. Error-Aware Imitation Learning pairs anomaly detection with recovery or termination in simulated mobile manipulation [67], illustrating why recoverable and unrecoverable deviations need separate outcomes.

B. REACTIVE AND ANTICIPATORY RECOVERY

Reactive recovery begins after evidence of deviation. Anticipatory approaches prepare branches or constraints before an incident; AgentChord is a recent example of agentic task-graph recovery [68]. Preparation can reduce planning delay, but it requires either identifying plausible failure branches or maintaining alternatives during execution. A cached plan needs its preconditions rechecked. B2FF selects a pre-imagined milestone as a fixed recovery condition and separately evaluates controlled entry and an online trigger [69]. This exposes both the recovery interface and its dependence on trigger timing.

FLARE combines augmented corrective behavior with object-centered reset skills and a supervisory choice of response [70]. Agentic reinforcement learning studies robust execution and recovery modes in simulated manipulation [71]. Their evidence differs in substrate and publication status. Repeatable simulated failure reconstruction contrasts with approximate physical reconstruction, which creates a transfer problem.

C. THE MONITOR AND RECOVERY POLICY MUST BE SELECTED TOGETHER

A detector with high recall may raise alarms in states from which its associated controller cannot recover. A conservative detector can preserve apparent recovery success by selecting only easy cases. Consequently, conditional recovery success among attempted repairs is insufficient. Let A denote the first monitor-triggered branch in an episode, and let V indicate that the permitted autonomous recovery remains viable at delivery. Under a protocol in which an alarmed, nonviable branch ends in stopping or human handoff, autonomous completion decomposes as

$$P(S_{auto}) = P(S_{auto} \cap A^c) + P(A)P(V \mid A)P(S_{auto} \mid A, V). \quad (4)$$

Equation (4) describes the first triggered branch; multiple alarms or repeated recovery cycles require an event-history formulation with conditional transitions and explicit stopping rules. The terms are conditional, so no independence is assumed. False alarms are included in A; direct completion includes success without a monitor-triggered repair. If the protocol allows success on a branch labeled nonviable, the definition of V or the decomposition must be expanded. This bookkeeping makes the main point explicit: better conditional repair performance need not increase total autonomous completion when alarm selection and delivery change.

A sound evaluation therefore compares policy–monitor–recovery combinations under matched budgets and event sets. It records unsuccessful and abandoned recovery attempts, as well as new hazards caused by repair. Repeated retry without a state change should have a declared stopping rule. A recovery controller also needs its own monitor: returning control prematurely can recreate the same failure, while an overly strict termination condition can trap the system in recovery.

IX. HUMAN INTERVENTION AND RECOVERY DATA

A. INTERVENTION IS BOTH A RESOURCE AND A SELECTION MECHANISM

Human help can take several forms: approving a plan, correcting language, labeling an outcome, teleoperating a segment, or physically resetting the scene. These interventions differ in latency, effort, observability, and what supervision they generate. Human-Gated Dataset Aggregation (HG-DAgger) sup-

TABLE 5. Detection and uncertainty methods: signals, supervision and interpretation

Method / family	Evidence and supervision	Decision supported	Limit of the extracted evidence
Ensembles / calibration [38], [52]	Predictor variation or held-out score calibration.	Uncertainty or calibrated predictive scores.	Not a direct guarantee of manipulation success.
Sentinel [32]	Action consistency plus semantic task-progress evidence.	Complementary runtime failure alarms.	Signal complementarity does not establish a universal intervention deadline.
FAIL-Detect [53]	Successful training data; learned/post-hoc scalar scores and sequential conformal monitoring.	Out-of-distribution runtime alarms without failure training examples.	Guarantees concern the stated calibration construction and assumptions.
Success VLM / AHA [55], [56]	Visual-language goal assessment or failure-oriented supervision.	Success judgment or interpretable failure feedback.	Semantic judgments are not verified physical causes.
KnowNo [59]	Calibrated uncertainty over language-planning alternatives.	Ask for help when planning uncertainty warrants it.	Plan-selection scope differs from low-level contact safety.
KnowLoop [60]	Uncertainty in closed-loop language planning.	Failure assessment during iterative planning.	Planner-level evidence must not be promoted to motor-level guarantees.
Latent safety filtering [61]	Learned latent uncertainty and filtering.	Avoidance of out-of-distribution actions/states.	Constraint avoidance and original-goal completion are distinct.
FailureSpot (preprint) [51]	Weak action-chunk supervision and targeted timestamp annotation.	Temporally localized failure detection.	Terminal episode labels cannot define every frame's onset correctly.
Physical / multimodal monitoring [57], [58]	Force/torque, tactile and kinematic streams; motor current plus dynamics.	Anomaly scores or collision alarms; nominal-motion training.	Physical contact evidence complements semantic state; validate task phase, sensor alignment and delivery time.

The last column gives interpretive boundaries, not a claim that every source explicitly analyzes the proposed operational metric. Detailed method descriptions are restricted to inspected source evidence.

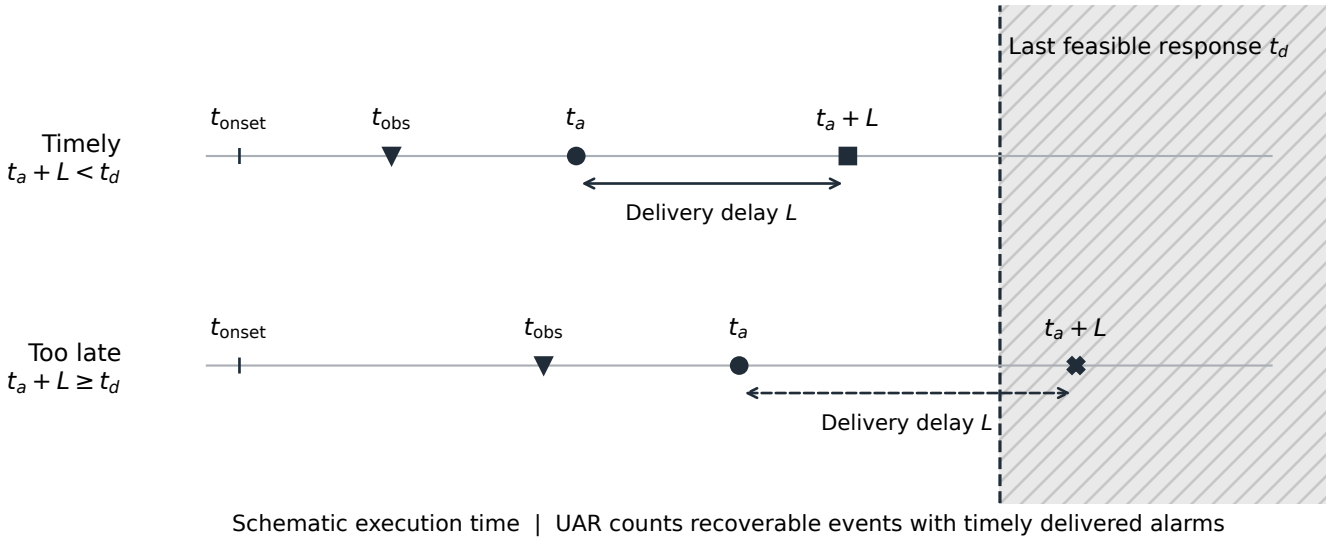


FIGURE 4. Intervention timing for Eq. (2). Observation and detection precede alarm emission; L is the remaining delivery delay. Only the timely path contributes to usable-alarm recall in Eq. (3).

plies a classic interactive-learning precedent, originally evaluated in driving rather than manipulation [74]. ThriftyDAgger considers how novelty and risk can gate interaction under a budget [75]. Their transferable lesson is that the query rule shapes operational assistance and the training distribution.

Separate active control, monitoring, labeling, reset, and waiting time to expose both intervention and availability costs. Report assisted and autonomous completion separately. The expected availability and competence of the human are part of the operating envelope in Eq. (1).

B. RECOVERY DEMONSTRATIONS REQUIRE TRANSITIONS, NOT JUST LABELS

A useful recovery record connects the preceding task context, failure onset, resulting state, selected intervention, corrective actions, and the resumed task outcome. A terminal failure label alone cannot specify how to return to useful support. RaC explicitly studies recovery and correction segments for long-horizon learning [25]. RESample explores augmentation through exploratory sampling and recovery behavior [72]. EgoRecovery investigates transfer from human recovery demonstrations [73]. FailSafe generates failure

TABLE 6. Recovery mechanisms compared by target and actuation path

Method / family	Trigger or state evidence	Repair mechanism / target	Transfer and evaluation boundary
Recovery RL [62]	Learned risk of constraint violation.	Separate task and recovery policies; return toward a safe region.	Contact-rich manipulation in simulation; physical obstacle avoidance is adjacent evidence.
REFLECT [63]	Summarized execution observations and failure explanation.	Use feedback to select corrective behavior.	Explanation quality and physical repair success are separate outcomes.
RePLan / DoReMi [64], [65]	Perception-language evidence or plan–execution mismatch.	Revise the plan or its grounded continuation.	Repair remains limited by available skills and state estimates.
RACER [24]	Rich language feedback about deviations.	Language-conditioned corrective policy.	Instruction usefulness depends on corrective training support.
RaC (preprint) [25]	Off-trajectory states and correction demonstrations.	Recover toward a familiar region, then correct and continue.	Failure coverage and resumed task success both matter.
FLARE (CVPR 2026) [70]	Supervisory response selection from execution evidence.	Perturbation/bridging augmentation plus object-centered reset skills.	Physical evidence includes two reported tasks; no cross-method pooled effect.
AgentChord (preprint) [68]	Task-graph reasoning and monitored execution.	Prepare anticipatory recovery branches.	Precomputed branches still require valid physical preconditions.
Agentic RL (preprint) [71]	Execution state with high-level recovery decisions.	Learn robust execution/recovery modes.	Extracted evidence is simulated manipulation.
Elastic querying / VLCP (preprints) [33], [66]	State difficulty or newly observed execution context.	Adjust query/chunk schedule or rewrite control code.	Execution adaptation differs from guaranteed safe repair; VLCP tests simulation.
RESample / EgoRecovery (preprints) [72], [73]	Exploration-based or human recovery examples.	Expand corrective state/action support.	Synthetic or human-source transfer requires explicit held-out failure tests.

Recovery targets include safe return, local correction, state restoration and plan repair. They are not interchangeable with successful completion of the original goal.

cases paired with executable corrections in simulation [76]. These collection procedures cover different state–action distributions. Transfer to naturally occurring failures requires deployment tests.

Synthetic perturbations provide control and scale, but a perturbed expert trajectory can omit the dynamics of an actual failed attempt. Human demonstrations contain strategy information, but cross-embodiment transfer must account for reachability, contact, sensor perspective, and action representation. A recovery dataset should therefore record the source of each state: natural policy failure, injected perturbation, human reconstruction, or simulator reset. Mixing these sources without provenance makes it difficult to explain why a method transfers or fails.

C. INTERVENTION DATA ARE NOT AN UNBIASED FAILURE CENSUS

Human visibility and correction preferences, or a monitor threshold, select which failures enter the intervention data. Let Q be the event that a trajectory segment is collected for correction. In general, $P(s \mid Q)$ differs from the deployed state distribution $P(s)$. Training on this selected data may improve selected failures while leaving silent or unobserved failures unchanged.

The appropriate response is to log query probabilities where defined, trigger reasons, suppressed alarms, and outcome attribution. Separate test sets should evaluate both corrected modes and unselected or newly emerging modes. Counterfactual claims require care: an intervention prevents observing the untreated future of the same physical episode. Randomized, simulation-based, or safely bounded comparisons can help, but an intervention log alone does not identify the causal benefit of help. RECALL directly studies uncertainty-guided recovery demonstration collection and finds an adaptation–retention tradeoff when fine-tuning on selected corrective data [77]. Simulator resets enable matched collection from saved states; physical replication requires additional control.

X. CONTINUAL AND ADAPTIVE POLICY IMPROVEMENT

A. DIFFERENT UPDATE MECHANISMS SHOULD NOT SHARE ONE LABEL

Table 7 distinguishes interactive imitation, reinforcement learning, autonomous reset infrastructure, value-guided inference, and sequential retention. Human-in-the-loop reinforcement learning demonstrates precise physical manipulation with corrective assistance [78]. Experience-based VLA learning, exemplified by $\pi^*_{0.6}$, incorporates demonstrations, autonomous experience, and interventions into policy improvement [26]. Retention requires testing earlier skills after each update.

Value guidance can improve action selection using a learned value function without the same kind of policy-weight update [79]. Reset-free or autonomous training systems reduce manual preparation between learning episodes. MEDAL++, asynchronous human feedback, and reuse of prior data address complementary parts of this infrastructure [80]–[82]. Fail2Progress uses observed failures to generate targeted simulated data for skill-effect-model updates [83]. These mechanisms improve learning support; retained skills and monitor validity require separate tests.

B. RETENTION IS PART OF THE OUTCOME

Catastrophic forgetting is a reason to evaluate updates against a stable earlier-task set [84]. The recent continual-learning survey already organizes task formulations, techniques, benchmarks, and retention challenges in foundation-model robotics [19]; the additional focus here is how failure selection and monitor changes affect those claims. The relevant unit may include objects, environments, embodiments, or failure-handling routines, not just named tasks. LIBERO explicitly targets lifelong transfer [41], requiring a sequential evaluation schedule beyond a new-task before–after score.

For task i, let $R_{i,j}$ be a declared reliability metric on a fixed test distribution after update j. A retention report can show the full matrix R and the change $R_{i,j} - R_{i,i}$ for earlier tasks, while preserving operational budgets. A mean over all tasks can conceal severe regression in a safety-critical minority. Recent evidence also shows why forgetting should be measured rather than presumed. Pretrained VLAs with experience replay can resist forgetting in simulated sequential tasks [85], while a real-world continual-learning study reports forgetting under naive fine-tuning and mitigation through configured replay [86]. PHASER studies phase-aware replay allocation [87]; Simple Recipe Works finds strong sequential adaptation with low-rank updates and on-policy reinforcement learning in its tested settings [88]. These results differ in update objective, parameterization, data allocation, and task stream. Recurrence should likewise be measured on stable failure categories and exposure denominators, rather than raw counts collected under changing monitor thresholds. Fewer logged failures may reflect weaker detection or fewer opportunities, not better control.

C. UPDATES ALSO INVALIDATE MONITORING ASSUMPTIONS

Policy updates change the monitor's inputs, action distribution, and timing, potentially invalidating calibrated scores. Changed chunk lengths or recovery behavior can also expose previously unseen states during longer autonomous episodes. Validation must therefore track the evolving system and its induced population.

A continual-improvement loop should version the policy, monitor, recovery controller, data-selection rule, and calibration set together. Candidate updates should be tested on preserved nominal and failure-state suites, then on an explicitly delimited physical envelope. Rollback should restore a coherent component set. These principles make improvement claims testable as interacting components evolve. They do not constitute a certification procedure.

XI. COMPARATIVE ANALYSIS AND NEW SYNTHESIS

A. WHY APPARENTLY CONFLICTING FINDINGS CAN COEXIST

Scale improves transfer, but does not remove recovery shift. Large policy and dataset studies expand nominal competence [4], [6], [29]; robustness and recovery benchmarks expose failures under changed conditions [22], [23]. These findings are compatible because their conditioning distributions differ. The key question is how scaling changes the composition and recoverability of residual failures.

Uncertainty can help without identifying every failure. Success-only outlier detection and semantic progress monitoring target different error structures [32], [53]. A detector can improve discrimination on its test distribution while missing confident plan errors or producing late alarms. The synthesis is to evaluate complementary signals against common event annotations, then select operating points with the paired recovery controller. Averaging unrelated area-under-curve values would not resolve the disagreement.

Recovery data can improve correction without guaranteeing ordinary execution gains. Corrective supervision enriches post-failure support [24], [25]; LIBERO-RECOVER reports limits in the transfer from recovery training to failures encountered during standard execution [23]. The relevant difference is the distribution of encountered states and the control loop through which a correction is delivered. Reconstructed and naturally induced failures can therefore yield different outcomes.

Autonomous improvement can coexist with external maintenance. Reset infrastructure and human-assisted reinforcement learning address different bottlenecks [78], [80]. An update may improve task execution while requiring a reliable reset controller, outcome labeling, or operator availability. Reporting the assistance channel and resource budget allows these contributions to be compared without dismissing useful hybrid systems or overstating autonomy.

Resistance to forgetting and correction-induced forgetting are compatible. Replay-supported pretraining results and on-policy adaptation studies [85], [88] do not contradict RECALL's deterioration under recovery-only updates [77]. Selected off-nominal examples alter the update distribution, whereas replay or constrained updates preserve different parts of prior behavior. Together with physical continual-learning evidence [86], these findings narrow the open problem from whether VLAs can learn sequentially to which failure-focused update regimes retain operational reliability and valid monitoring.

B. AN ILLUSTRATIVE QUANTITATIVE REANALYSIS

To show how small physical evaluations should be interpreted, Fig. 5 reanalyzes the two real-world tasks in FLARE's Table 2 [70]. The table specifies 40 trials and reports baseline versus FLARE success rates of 62.5% versus 75.0% for stacking three blocks, and 45.0% versus 55.0% for inserting a U-shaped block. Interpreting the stated trial count as applying to each method–task condition gives integer success counts of 25, 30, 18, and 22 out of 40. That interpretation is stated explicitly because the source supplies rates rather than per-trial records.

For each condition, 95% Wilson score intervals were computed under the stated independent-binomial assumptions; the formula and inputs are retained in the supplement and worksheet. Under this independent-binomial illustration, the

TABLE 7. Learning and assistance mechanisms: what improves and what must still be evaluated

Mechanism / examples	Data or update path	Demonstrated research role	Required separate assessment
DAgger / HG-DAgger [35], [74]	Expert labels or human-gated corrections.	Address learner-induced distribution shift; HG-DAgger's original domain is driving.	Manipulation transfer and assistance selection bias.
ThriftyDAgger [75]	Budget-aware novelty/risk gating for interaction.	Allocate expert help in interactive imitation learning.	Active help, observation and reset effort separately.
HIL reinforcement learning [78]	Human corrections plus robot reinforcement learning.	Precise physical manipulation through interactive learning.	Retention of previous tasks and cost of ongoing assistance.
Experience-based VLA (preprint) [26]	Demonstrations, autonomous experience and interventions.	Policy improvement using experience and feedback.	Measure earlier-task retention after each update.
Autonomous training infrastructure [80]–[82]	Learned reset/return, asynchronous feedback or prior data.	Reduce manual bottlenecks in real-world learning.	Reset cost, evaluator error and original-task recovery.
Value guidance [79]	Learned value estimates guide candidate actions.	Improve generalist action selection at inference.	Distinguish guidance from policy-weight adaptation.
Forgetting / lifelong evaluation [41], [84]	Sequential task updates and earlier-task assessment.	Motivate retention and transfer measurements.	Use stable task/failure distributions and operational budgets.
Pretraining and replay [85], [86]	Sequential VLA tasks with replay; simulated and physical studies.	Retention depends on pretraining and update/data configuration.	Test retention under the intended update and data regime.
Selected recovery / phase replay (preprints) [77], [87]	Uncertainty-guided correction or phase-aware memory allocation.	Study adaptation–retention tradeoffs and replay selection.	Match collection effort, memory and task exposure.
Continual on-policy RL (preprint) [88]	Low-rank adaptation and sequential reinforcement learning.	Strong simple fine-tuning under the tested update regime.	Compare explicitly with recovery-only imitation updates.

Rows compare learning mechanisms rather than rank effectiveness. Continual improvement requires a sequence of evaluations, including prior capabilities and monitoring validity.

point differences are 12.5 and 10 percentage points, with broad condition-wise intervals. Interval overlap establishes neither paired significance nor equivalence. Missing trial pairing, session dependence, and task-sampling information preclude a paired treatment-effect interval.

This formally published example was selected for its explicit physical-trial denominator and relevant recovery framework, rather than through meta-analysis screening. Its two tasks illustrate uncertainty in reported gains; broader failure coverage requires additional tests. The CVPR version confirms the same source rates and trial-count wording; the worksheet preserves all reconstruction assumptions.

Figure 6 summarizes the coded evidence rather than ranking effectiveness. Of the 51 coded studies, 36 contain simulated and 31 physical evidence (overlapping counts), while four meet the declared sustained-operation rule: AutoEval, MEDAL++, HIL-SERL, and prior-data autonomous RL [48], [78], [80], [82]. Their accounts quantify reset frequency or intervention burden during evaluation or training. RECAP reports long runs and throughput, but lacks assistance costs on the same time axis [26]. Thus physical evidence and long task horizons alone do not supply a complete account of operational reliability across updates. This corpus-specific gap motivates joining timing, reset, assistance, and retained-monitor measurements; it is not a prevalence estimate for all robotics research.

C. A MINIMUM RECOVERY-CONDITIONED REPORT

Table 8 presents our proposed *Minimum Reporting Framework for Recovery-Conditioned Reliability*. It operationalizes the interfaces in Fig. 1: Eq. (2) defines response feasibility, Eq. (3) retains missed and late alarms in the event population, and Eq. (4) links branch selection to autonomous completion. Its most important distinction is between a detector-conditioned denominator and the complete event population. Report recovery success both among attempted repairs and among all eligible failures. The former describes controller execution on selected states; the latter exposes missed alarms and nonviable delivery. Report successful original-goal completion separately from merely reaching a safe or reset state.

Recovery cost also matters: time-to-resume, goals per operating hour, human effort, and constraint violations complement episode success. Compare these outcomes under common budgets; application-specific consequences make a universal weighted score inappropriate.

Supplementary Table S1 applies these fields to Sentinel, FailureSpot, LIBERO-RECOVER, FLARE, and B2FF [23], [32], [51], [69], [70]. The contrasts identify complementary missing links: Sentinel reports detection timing but no paired repair deadline; FailureSpot annotates onset and recovered success without measuring delivered intervention; LIBERO-RECOVER measures original-goal recovery from constructed starts without an online alarm population. FLARE reports integrated task outcomes but not paired event timing. B2FF reports a separate online trigger, runtime costs, and controlled delay sensitivity, but lacks per-event viable deadlines paired with delivered responses and missed events. UAR is therefore *not computable from reported evidence* in these five inspected cases. This is a reporting result, not a performance ranking. The comparison shows why success rates and detector metrics cannot substitute for the joined event record required by Table 8.

XII. OPEN CHALLENGES AND RESEARCH DIRECTIONS

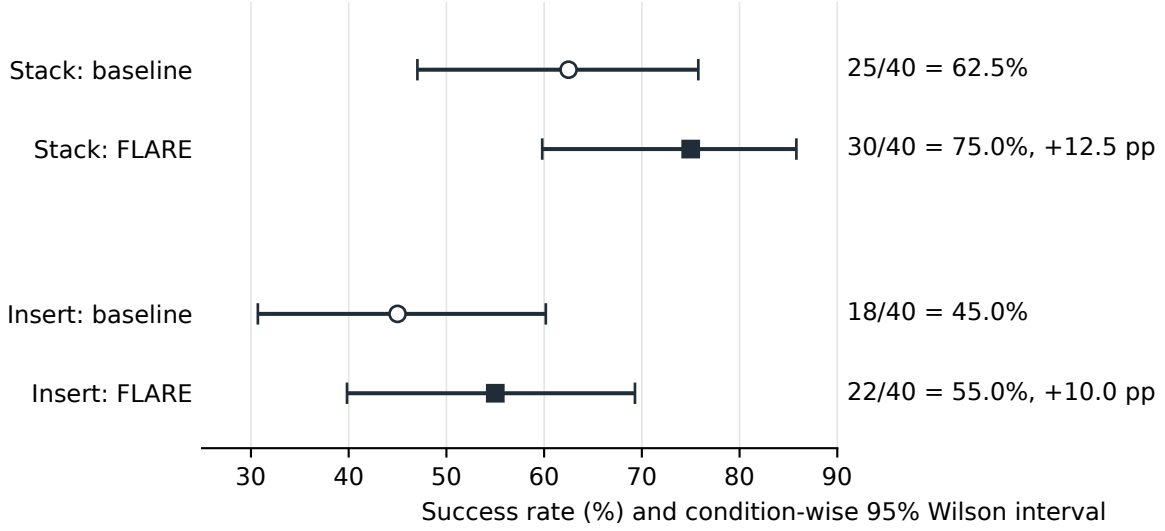


FIGURE 5. **FLARE Table 2 reanalysis [70]. Points reproduce published rates; counts assume 40 trials per method–task condition. Bars are condition-wise 95% Wilson intervals under an independent-binomial model. Point differences are descriptive, not paired significance tests.**

51 coded studies from 88 included works

	Component	Simulation	Physical	Sustained operation
Robustness	**1** 1 / 0	**11** 7 / 4	**6** 5 / 1	**0** 0 / 0
Detection	**5** 2 / 3	**4** 3 / 1	**3** 3 / 0	**0** 0 / 0
Recovery	**0** 0 / 0	**16** 8 / 8	**16** 9 / 7	**3** 3 / 0
Human intervention	**0** 0 / 0	**2** 1 / 1	**7** 5 / 2	**2** 2 / 0
Continual / adaptive improvement	**0** 0 / 0	**11** 7 / 4	**9** 7 / 2	**3** 3 / 0

Cell: total works; below: peer-reviewed / preprint. Cells overlap; uncoded evidence is not zero prevalence.

FIGURE 6. **Programmatically generated evidence map from 51 coded studies within the 88-work corpus. Each cell counts distinct works, with formal/preprint counts below. Domain–substrate codes overlap. Thirty-four contextual works and three restricted-extraction records are excluded. Physical monitoring can use recorded rollouts; sustained operation additionally requires duration plus quantified reset or assistance frequency/cost.**

A. TIMELY DETECTION: RECOVERY-CONDITIONED MONITORING

Contact failures can develop faster than semantic reasoning can respond. Multimodal anomaly and collision detectors [57], [58] offer physical signals that complement the semantic and action-consistency signals in Sentinel and FAIL-Detect [32], [53]. The open problem is to choose sensing, alarm thresholds, and action cancellation jointly with recoverable operating regions. A testable study would hold the disturbance distribution fixed, vary observation availability and delivery delay, and compare monitors selected by discrimination scores with monitors selected by usable-alarm recall and autonomous completion. Missing sensors, synchronization errors, and embodiment-specific calibration should be explicit test conditions.

B. FAILURE COVERAGE: PROVENANCE-AWARE FAILURE SUITES

Failures combine objects, dynamics, instructions, preceding actions, and responses. Random perturbations can overrepresent easy-to-generate cases, whereas operational logs select the failures encountered and detected by a particular system. Factorized evaluation [36], [49] and policy-generated failure benchmarks [23] address complementary parts of this cov-

TABLE 8. Proposed Minimum Reporting Framework for Recovery-Conditioned Reliability

Quantity / record	Denominator or conditioning	Why it is needed	Essential companion information
Autonomous completion	All initiated episodes; fixed budget; no human takeover.	Measures original-goal completion.	Constraints, retries, timeout and assistance definition.
Attempt / episode success	Attempts versus episodes with declared retries.	Prevents retry allowance from hiding weak execution.	Number of attempts and per-episode stopping rule.
Usable-alarm recall	All eligible recoverable failure events.	Counts missed and late alarms.	Failure onset, response deadline and latency definition.
False-alarm burden	Alarms on eligible nonfailure exposure / operating hour.	Captures unnecessary interruption cost.	Label policy, paused time and recovery invoked.
Conditional recovery success	Successful repairs / all attempted repairs.	Describes selected controller performance.	Trigger policy and failure-state distribution.
Failure-level recovery	Successful original-goal continuation / all eligible failures.	Includes failures not selected for repair.	Unrecoverable cases, missed alarms and human handoffs.
Time-to-resume	Elapsed time from declared event to validated handback.	Measures repair delay, not just eventual success.	Sensing, compute, cancellation and actuation components.
Sustained throughput	Completed original goals / elapsed operating time.	Includes reset and downtime.	Maintenance, human availability and task mix.
Assistance effort	Control, labeling, observation, reset and waiting time.	Preserves different human contributions.	Operator protocol and supervision channel.
Retention matrix	Same earlier-task tests after successive updates.	Detects regression hidden by new-task improvement.	Task ordering, seeds, budgets and component versions.
Failure recurrence	Repeated failure class / matched exposure.	Distinguishes learning from changed collection.	Stable monitor/labels or explicit recalibration.
Constraint violations	Events and severity under a declared envelope.	Separates safety from task completion.	Near misses, termination and recovery-induced violations.

Proposed by this survey. Report the listed quantities for the coupled policy, monitor and recovery system under matched budgets and explicit event eligibility.

erage problem. Linked nominal, perturbed, and post-failure suites should record the originating policy, intervention rule, and reconstruction method. Splits across policy families and correction sources would test transfer beyond training errors. Matched simulated and physical cases could reveal which mechanisms transfer. Evaluation effort should be reported separately for common errors and rare high-consequence events, with the sampled suite distinguished from the deployment distribution.

C. GROUNDED DIAGNOSIS: VERIFIABLE REPAIR

Natural-language explanations support communication, but an observation such as an absent object leaves several causal hypotheses open. A repair interface should attach a checkable target to each proposed response: restored support, satisfied preconditions, safe containment, or original-goal completion. REFLECT and language-conditioned correction [24], [63] motivate testing diagnosis and action together. A controlled comparison could hold the low-level skills fixed and vary evidence history or explanation quality, measuring verified physical restoration and recurrence. Cases with occlusion, slip, and missing skills would separate observation, diagnosis, and actuation bottlenecks. Verification of a symbolic relation should be paired with contact-state evidence when the task depends on stable interaction.

D. SELECTED CORRECTION DATA: SELECTION-AWARE EVALUATION

Interactive imitation and recovery-data methods [25], [73], [75] select where supervision enters the loop. A gain may arise from improved actions, easier selected events, or additional assistance. Future comparisons should match active human effort, document the selection rule, and evaluate unqueried failures as well as attempted repairs. Language feedback, teleoperation, and physical reset provide different information with different delays. Recording observation, control, labeling, waiting, and maintenance separately makes those channels comparable. Recovery-only updates should also be tested against nominal behavior and earlier failure-handling capabilities, following the adaptation–retention tension exposed by RECALL [77].

E. CONTINUAL IMPROVEMENT: VERSIONED SYSTEM EVALUATION

The corpus map locates the missing operational evidence, while Supplementary Table S1 identifies missing event links needed to interpret timing and recovery together. The long-horizon review similarly identifies evaluation gaps [18]. Experience-based learning and autonomous training infrastructure [26], [80], [82] motivate studies that release a reliability matrix across earlier and new tasks, preserve failure probes, and revalidate the monitor after each update. The testable hypothesis is that component improvements compose only when their interfaces remain valid. Event logs, evaluator definitions, calibration sets, policy and recovery versions, and rollback outcomes would expose regressions and distinguish reduced recurrence from changed observation or selection.

F. CONTAINMENT AND RESTORATION: SUSTAINED SERVICE VALUE

Lifecycle risk-control surveys already motivate post-incident response [12], [15]. A manipulation system additionally needs evidence connecting containment and restoration to useful operation. Studies should declare when stop or human handoff is preferred, measure the resources required, and report throughput and downtime alongside task completion. Safe refusal merits containment credit; autonomous goal completion has its own denominator. The architecture in Fig. 1 and reporting framework in Table 8 make these outcomes traceable across evaluation, response, and learning.

XIII. CONCLUSION

Reliable generalist manipulation depends on the full chain from failure occurrence through observable evidence, timely detection, recoverability, and recovery outcome to continual improvement. The failure-conditioned evidence framework in Fig. 1 makes the monitor–recovery–learning interfaces explicit. Alarms must arrive while a permitted response remains viable; recovery rates must account for missed and unselected failures; and adaptation must retain earlier capabilities and valid monitoring. The proposed operational metric UAR addresses timely alarm delivery, while the Proposed Minimum Reporting Framework in Table 8 consolidates timing, selection, completion, and recurrence requirements. The corpus evidence map and five-study reporting comparison locate gaps between available measurements and complete reliability claims. Together, these contributions support comparison under declared conditions while preserving distinctions between substrates and publication status.

ACKNOWLEDGMENT

The author retains responsibility for study selection, research judgments, and all manuscript claims. No new robot experiment was conducted. This research received no funding. The author declares no conflicts of interest.

REFERENCES

[1] A. Brohan *et al.*, "RT-1: Robotics Transformer for Real-World Control at Scale," in *Robotics: Science and Systems XIX*, vol. 19, 2023. [Online]. Available: https://roboticsproceedings.org/rss19/p025.html

[2] B. Zitkovich *et al.*, "RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 2165–2183. [Online]. Available: https://proceedings.mlr.press/v229/zitkovich23a.html

[3] D. Ghosh *et al.*, "Octo: An Open-Source Generalist Robot Policy," in *Robotics: Science and Systems XX*, vol. 20, 2024. [Online]. Available: https://roboticsproceedings.org/rss20/p090.html

[4] M. J. Kim *et al.*, "OpenVLA: An Open-Source Vision-Language-Action Model," in *Proceedings of The 8th Conference on Robot Learning*, vol. 270, 2025, pp. 2679–2713. [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[5] K. Black *et al.*, "π_0: A Vision-Language-Action Flow Model for General Robot Control," in *Robotics: Science and Systems XXI*, vol. 21, 2025. [Online]. Available: https://roboticsproceedings.org/rss21/p010.html

[6] ——, "$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization," in *Proceedings of The 9th Conference on Robot Learning*, vol. 305, 2025, pp. 17–40. [Online]. Available: https://proceedings.mlr.press/v305/black25a.html

[7] Physical Intelligence *et al.*, "$\pi_{0.7}$: a Steerable Generalist Robotic Foundation Model with Emergent Capabilities," arXiv preprint, arXiv:2604.15483, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2604.15483

[8] Y. Hu *et al.*, "Toward General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis," arXiv preprint, arXiv:2312.08782, 2023, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2312.08782

[9] Z. Xu *et al.*, "A Survey on Robotics with Foundation Models: toward Embodied AI," arXiv preprint, arXiv:2402.02385, 2024, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2402.02385

[10] D. Li *et al.*, "What foundation models can bring for robot learning in manipulation: A survey," *The International Journal of Robotics Research*, vol. 45, no. 7, pp. 1091–1142, 2026. [Online]. Available: https://doi.org/10.1177/02783649251390579

[11] S. Bai *et al.*, "Embodied Robot Manipulation in the Era of Foundation Models: Planning and Learning Perspectives," arXiv preprint, arXiv:2512.22983, 2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2512.22983

[12] T. Kojima *et al.*, "A Comprehensive Survey on Physical Risk Control in the Era of Foundation Model-enabled Robotics," in *Thirty-Fourth International Joint Conference on Artificial Intelligence*, 2025, pp. 10 517–10 527. [Online]. Available: https://www.ijcai.org/proceedings/2025/1168

[13] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, "Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications," *IEEE Access*, vol. 13, pp. 162 467–162 504, 2025. [Online]. Available: https://doi.org/10.1109/access.2025.3609980

[14] Y. Ma, Z. Song, Y. Zhuang, J. Hao, and I. King, "A Survey on Vision–Language–Action Models for Embodied AI," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 37, no. 7, pp. 3031–3051, 2026. [Online]. Available: https://doi.org/10.1109/tnnls.2025.3650584

[15] Q. Li *et al.*, "Vision-Language-Action Safety: Threats, Challenges, Evaluations, and Mechanisms," arXiv preprint, arXiv:2604.23775, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2604.23775

[16] Z. Wang *et al.*, "Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines," arXiv preprint, arXiv:2604.23001, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2604.23001

[17] Z. Song *et al.*, "The Evaluation Bottleneck of Vision-Language-Action Models: A Evaluation-Centric Survey," Preprints.org preprint, 2026, preprint; accessed Sep. 10, 2026. [Online]. Available: https://www.preprints.org/manuscript/202606.0425

[18] M. Acs and X. Zhong, "A scoping review of reinforcement learning methods for long horizon robotic manipulation," *Discover Robotics*, vol. 2, no. 1, 2026, art. no. 22. [Online]. Available: https://doi.org/10.1007/s44430-026-00037-1

[19] Z. Mai *et al.*, "A Survey of Continual Learning for Robotics in the Foundation Model Era," TechRxiv preprint, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://doi.org/10.36227/techrxiv.176972367.76460794/v2

[20] W. Yuan *et al.*, "Safety of Vision-Language-Action Models: A Survey from Lifecycle Perspectives," Authorea preprint, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://doi.org/10.22541/au.177524426.60806944/v1

[21] X. Zhou *et al.*, "LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization," arXiv preprint, arXiv:2510.03827, 2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2510.03827

[22] G. Wang *et al.*, "LIBERO-X: Robustness Litmus for Vision-Language-Action Models," in *Robotics: Science and Systems XXII*, vol. 22, 2026. [Online]. Available: https://www.roboticsproceedings.org/rss22/p097.html

[23] L. Liu *et al.*, "LIBERO-RECOVER: Beyond Task Success Towards Failure Recovery in Robotic Manipulation Models," arXiv preprint, arXiv:2609.05178, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2609.05178

[24] Y. Dai, J. Lee, N. Fazeli, and J. Chai, "RACER: Rich Language-Guided Failure Recovery Policies for Imitation Learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 15 657–15 664. [Online]. Available: https://doi.org/10.1109/icra55743.2025.11127799

[25] Z. Hu *et al.*, "RaC: Robot Learning for Long-Horizon Tasks by Scaling Recovery and Correction," arXiv preprint, arXiv:2509.07953,

2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2509.07953

[26] Physical Intelligence *et al.*, "$\pi^*_{0.6}$: a VLA That Learns From Experience," arXiv preprint, arXiv:2511.14759, 2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2511.14759

[27] C. Chi *et al.*, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10-11, pp. 1684–1704, 2025. [Online]. Available: https://doi.org/10.1177/02783649241273668

[28] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Robotics: Science and Systems XIX*, vol. 19, 2023. [Online]. Available: https://www.roboticsproceedings.org/rss19/p016.html

[29] A. O'Neill *et al.*, "Open X-Embodiment: Robotic Learning Datasets and RT-X Models : Open X-Embodiment Collaboration," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 6892–6903. [Online]. Available: https://doi.org/10.1109/icra57147.2024.10611477

[30] H. R. Walke *et al.*, "BridgeData V2: A Dataset for Robot Learning at Scale," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 1723–1736. [Online]. Available: https://proceedings.mlr.press/v229/walke23a.html

[31] A. Khazatsky *et al.*, "DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset," in *Robotics: Science and Systems XX*, vol. 20, 2024. [Online]. Available: https://roboticsproceedings.org/rss20/p120.html

[32] C. Agia *et al.*, "Unpacking Failure Modes of Generative Policies: Runtime Monitoring of Consistency and Progress," in *Proceedings of The 8th Conference on Robot Learning*, vol. 270, 2025, pp. 689–723. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[33] G. Wang *et al.*, "Elastic Queries Reinforcement Learning: Self-Aware Policy Execution for VLA Models," arXiv preprint, arXiv:2606.14375, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2606.14375

[34] B. Ichter *et al.*, "Do As I Can, Not As I Say: Grounding Language in Robotic Affordances," in *Proceedings of The 6th Conference on Robot Learning*, vol. 205, 2023, pp. 287–318. [Online]. Available: https://proceedings.mlr.press/v205/ichter23a.html

[35] S. Ross, G. Gordon, and D. Bagnell, "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, vol. 15, 2011, pp. 627–635. [Online]. Available: https://proceedings.mlr.press/v15/ross11a.html

[36] J. Gao, S. Belkhale, S. Dasari, A. Balakrishna, D. Shah, and D. Sadigh, "A Taxonomy for Evaluating Generalist Robot Manipulation Policies," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3182–3189, 2026. [Online]. Available: https://doi.org/10.1109/lra.2026.3656785

[37] D. Goel *et al.*, "Geometric Red-Teaming for Robotic Manipulation," in *Conference on Robot Learning*, vol. 305, 2025, pp. 41–67. [Online]. Available: https://proceedings.mlr.press/v305/goel25a.html

[38] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proceedings of the 34th International Conference on Machine Learning*, vol. 70, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[39] A. N. Angelopoulos and S. Bates, "Conformal Prediction: A Gentle Introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023. [Online]. Available: https://doi.org/10.1561/2200000101

[40] O. Mees, L. Hermann, E. Rosete-Beas, and W. B. Burgard, "CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7327–7334, 2022. [Online]. Available: https://doi.org/10.1109/lra.2022.3180108

[41] B. Liu *et al.*, "LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 44 776–44 791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html

[42] S. Nasiriany *et al.*, "RoboCasa: Large-Scale Simulation of Household Tasks for Generalist Robots," in *Robotics: Science and Systems XX*, vol. 20, 2024. [Online]. Available: https://roboticsproceedings.org/rss20/p050.html

[43] S. Nasiriany, S. Nasiriany, A. Maddukuri, and Y. Zhu, "RoboCasa365: A Large-Scale Simulation Framework for Training and Benchmarking Generalist Robots," arXiv preprint, arXiv:2603.04356, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2603.04356

[44] W. Pumacay, I. Singh, J. Duan, R. Krishna, J. Thomason, and D. Fox, "THE COLOSSEUM: A Benchmark for Evaluating Generalization for Robotic Manipulation," in *Robotics: Science and Systems XX*, vol. 20, 2024. [Online]. Available: https://www.roboticsproceedings.org/rss20/p133.html

[45] E. Zhao *et al.*, "ManipBench: Benchmarking Vision-Language Models for Low-Level Robot Manipulation," in *Proceedings of The 9th Conference on Robot Learning*, vol. 305, 2025, pp. 3413–3462. [Online]. Available: https://proceedings.mlr.press/v305/zhao25a.html

[46] Z. Navasardyan, T. Danielyan, and H. Davtyan, "FailBench: How Reliable are VLMs at Judging Robot Task Success?" arXiv preprint, arXiv:2609.03611, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2609.03611

[47] X. Li *et al.*, "Evaluating Real-World Robot Manipulation Policies in Simulation," in *Proceedings of The 8th Conference on Robot Learning*, vol. 270, 2025, pp. 3705–3728. [Online]. Available: https://proceedings.mlr.press/v270/li25c.html

[48] Z. Zhou, P. Atreya, Y. L. Tan, K. Pertsch, and S. Levine, "AutoEval: Autonomous Evaluation of Generalist Robot Manipulation Policies in the Real World," in *Proceedings of The 9th Conference on Robot Learning*, vol. 305, 2025, pp. 1997–2017. [Online]. Available: https://proceedings.mlr.press/v305/zhou25a.html

[49] A. Liao, H. Cui, K. Desingh, and A. Deshwal, "Active Real-World Factor-Based Evaluation for Generalist Robot Policies," arXiv preprint, arXiv:2607.14439, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2607.14439

[50] T. Jiang, X. Tan, S. Wheeler, L. Sun, T. W. Ayalew, and M. Walter, "What Are We Actually Benchmarking in Robot Manipulation?" arXiv preprint, arXiv:2606.04233, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2606.04233

[51] J. Ma, Z. Liu, and Y. Zhu, "FailureSpot: Label-Efficient Timestamp-Level Failure Detection for Vision-Language-Action Models," arXiv preprint, arXiv:2609.04277, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2609.04277

[52] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html

[53] C. Xu *et al.*, "Can We Detect Failures Without Failure Data? Uncertainty-Aware Runtime Failure Detection for Imitation Learning Policies," in *Robotics: Science and Systems XXI*, vol. 21, 2025. [Online]. Available: https://www.roboticsproceedings.org/rss21/p073.html

[54] Q. Gu *et al.*, "SAFE: Multitask Failure Detection for Vision-Language-Action Models," in *Advances in Neural Information Processing Systems*, vol. 38, 2026, pp. 40 041–40 076. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/392d0d05e2f514063e6ce6f8b370834c-Abstract-Conference.html

[55] Y. Du *et al.*, "Vision-Language Models as Success Detectors," in *Proceedings of The 2nd Conference on Lifelong Learning Agents*, vol. 232, 2023, pp. 120–136. [Online]. Available: https://proceedings.mlr.press/v232/du23b.html

[56] J. Duan *et al.*, "AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in Robotic Manipulation," in *International Conference on Learning Representations*, vol. 2025, 2025, pp. 45 493–45 517. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/70a06501001e1820fd1eb9ee821302d2-Abstract-Conference.html

[57] L. Yang, W. Yan, and H. Wu, "Comparison of deep learning-based methods in multimodal anomaly detection: A case study in human–robot collaboration," *Science Progress*, vol. 104, no. 2, 2021, art. no. 00368504211021192. [Online]. Available: https://doi.org/10.1177/00368504211021192

[58] K. M. Park, Y. Park, S. Yoon, and F. C. Park, "Collision Detection for Robot Manipulators Using Unsupervised Anomaly Detection Algorithms," *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 5, pp. 2841–2851, 2022. [Online]. Available: https://doi.org/10.1109/tmech.2021.3119057

[59] A. Z. Ren *et al.*, "Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 661–682. [Online]. Available: https://proceedings.mlr.press/v229/ren23a.html

[60] Z. Zheng, Q. Feng, H. Li, A. Knoll, and J. Feng, "Evaluating Uncertainty-based Failure Detection for Closed-Loop LLM Planners," arXiv preprint, arXiv:2406.00430, 2024, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2406.00430

[61] J. Seo, K. Nakamura, and A. Bajcsy, "Uncertainty-aware Latent Safety Filters for Avoiding Out-of-Distribution Failures," in *Proceedings of The 9th Conference on Robot Learning*, vol. 305, 2025, pp. 4442–4472. [Online]. Available: https://proceedings.mlr.press/v305/seo25a.html

[62] B. Thananjeyan *et al.*, "Recovery RL: Safe Reinforcement Learning With Learned Recovery Zones," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4915–4922, 2021. [Online]. Available: https://doi.org/10.1109/lra.2021.3070252

[63] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing Robot Experiences for Failure Explanation and Correction," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 3468–3484. [Online]. Available: https://proceedings.mlr.press/v229/liu23g.html

[64] M. Skreta, Z. Zhou, J. L. Yuan, K. Darvish, A. Aspuru-Guzik, and A. Garg, "RePLan: Robotic Replanning with Perception and Language Models," arXiv preprint, arXiv:2401.04157, 2024, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2401.04157

[65] Y. Guo, Y.-J. Wang, L. Zha, and J. Chen, "DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 12 124–12 131. [Online]. Available: https://doi.org/10.1109/iros58592.2024.10802284

[66] D. Naouali, M. Wu, C. Wong, A. Puthran, and O. G. Younis, "VLCP: Vision Language Control Policy Closed-Loop Code Replanning for Robot Manipulation," arXiv preprint, arXiv:2608.16978, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2608.16978

[67] J. Wong *et al.*, "Error-Aware Imitation Learning from Teleoperation Data for Mobile Manipulation," in *Conference on Robot Learning*, vol. 164, 2022, pp. 1367–1378. [Online]. Available: https://proceedings.mlr.press/v164/wong22a.html

[68] S. Xu *et al.*, "From Reaction to Anticipation: Proactive Failure Recovery through Agentic Task Graph for Robotic Manipulation," arXiv preprint, arXiv:2605.11951, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2605.11951

[69] S. Shin *et al.*, "Back to the Familiar Future: Failure Recovery for VLA Policies via Pre-Imagined Milestone Selection," arXiv preprint, arXiv:2606.09258, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2606.09258

[70] G. Zhao, Z. Tang, X. Chen, Z. Kuang, Y. Tian, and G. Li, "FLARE: A Failure-Aware Framework for Autonomous Correction and Recovery in Visual-Language Robotic Manipulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 22 391–22 401. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Zhao_FLARE_A_Failure-Aware_Framework_for_Autonomous_Correction_and_Recovery_in_CVPR_2026_paper.html

[71] X. Zhang, Y. Weng, Q. Liu, Y. Mu, and Y. Li, "Learning Robust Execution in Robotic Manipulation with Agentic Reinforcement Learning," arXiv preprint, arXiv:2607.13818, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2607.13818

[72] Y. Xue *et al.*, "RESample: A Robust Data Augmentation Framework via Exploratory Sampling for Robotic Manipulation," arXiv preprint, arXiv:2510.17640, 2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2510.17640

[73] Z. Ge *et al.*, "EgoRecovery: Acquiring Failure Recovery Ability Through Human Recovery Demonstration," arXiv preprint, arXiv:2607.19745, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2607.19745

[74] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer, "HG-DAgger: Interactive Imitation Learning with Human Experts," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 8077–8083. [Online]. Available: https://doi.org/10.1109/icra.2019.8793698

[75] R. Hoque, A. Balakrishna, E. Novoseller, A. Wilcox, D. S. Brown, and K. Goldberg, "ThriftyDAgger: Budget-Aware Novelty and Risk Gating for Interactive Imitation Learning," in *Proceedings of the 5th Conference on Robot Learning*, vol. 164, 2022, pp. 598–608. [Online]. Available: https://proceedings.mlr.press/v164/hoque22a.html

[76] Z. Lin *et al.*, "FailSafe: Reasoning and Recovery from Failures in Vision-Language-Action Models," arXiv preprint, arXiv:2510.01642, 2025, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2510.01642

[77] U. B. Karli and T. Fitzgerald, "RECALL: Recovery Experience Collection for Active Lifelong Learning in Vision-Language-Action Models," arXiv preprint, arXiv:2606.23617, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2606.23617

[78] J. Luo, C. Xu, J. Wu, and S. Levine, "Precise and dexterous robotic manipulation via human-in-the-loop reinforcement learning," *Science Robotics*, vol. 10, no. 105, 2025, art. no. eads5033. [Online]. Available: https://doi.org/10.1126/scirobotics.ads5033

[79] M. Nakamoto, O. Mees, A. Kumar, and S. Levine, "Steering Your Generalists: Improving Robotic Foundation Models via Value Guidance," in *Proceedings of The 8th Conference on Robot Learning*, vol. 270, 2025, pp. 4996–5013. [Online]. Available: https://proceedings.mlr.press/v270/nakamoto25a.html

[80] A. Sharma, A. M. Ahmed, R. Ahmad, and C. Finn, "Self-Improving Robots: End-to-End Autonomous Visuomotor Reinforcement Learning," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 3292–3308. [Online]. Available: https://proceedings.mlr.press/v229/sharma23b.html

[81] M. Balsells, M. T. Villasevil, Z. Wang, S. Desai, P. Agrawal, and A. Gupta, "Autonomous Robotic Reinforcement Learning with Asynchronous Human Feedback," in *Proceedings of The 7th Conference on Robot Learning*, vol. 229, 2023, pp. 774–799. [Online]. Available: https://proceedings.mlr.press/v229/balsells23a.html

[82] H. R. Walke *et al.*, "Don't Start From Scratch: Leveraging Prior Data to Automate Robotic Reinforcement Learning," in *Proceedings of The 6th Conference on Robot Learning*, vol. 205, 2023, pp. 1652–1662. [Online]. Available: https://proceedings.mlr.press/v205/walke23a.html

[83] Y. Huang, N. Alvina, M. D. Shanthi, and T. Hermans, "Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference," in *Conference on Robot Learning*, vol. 305, 2025, pp. 5581–5605. [Online]. Available: https://proceedings.mlr.press/v305/huang25d.html

[84] J. Kirkpatrick *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017. [Online]. Available: https://doi.org/10.1073/pnas.1611835114

[85] H. Liu, C. Kim, B. Liu, M. Liu, and Y. Zhu, "Pretrained Vision-Language-Action Models are Surprisingly Resistant to Forgetting in Continual Learning," in *International Conference on Machine Learning*, 2026. [Online]. Available: https://rpl.cs.utexas.edu/publications/2026/07/19/liu-icml26-continual/

[86] J. Zhu *et al.*, "Can Vision-Language-Action Models Learn from Real-World Data Continually without Forgetting?" arXiv preprint, arXiv:2605.26820, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2605.26820

[87] Z. Chen *et al.*, "PHASER: Phase-Aware and Semantic Experience Replay for Vision-Language-Action Models," arXiv preprint, arXiv:2606.03598, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2606.03598

[88] J. Hu *et al.*, "Simple Recipe Works: Vision-Language-Action Models are Natural Continual Learners with Reinforcement Learning," arXiv preprint, arXiv:2603.11653, 2026, preprint; accessed Sep. 16, 2026. [Online]. Available: https://arxiv.org/abs/2603.11653

XIONGYULE ZHOU is pursuing the B.Eng. degree in mechatronic engineering at Beijing Jiaotong University, Beijing, China, with expected completion in 2027. His university projects include learning-based robotic manipulation in simulation, vision-guided grasping and object handoff, and autonomous robot perception and control. His internship experience includes ROS-based bimanual teleoperation and demonstration-based learning, robot calibration and hardware–software synchronization, embodied-AI simulation, and humanoid motion-planning and control pipelines. His research interests include robot learning, reinforcement learning, sim-to-real transfer, robotic manipulation, and autonomous systems.